RefSel: Risk-Controlled Foreign-Key Discovery with Bounded Language-Model Evidence

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Abstract—Foreign keys specify how the tables of a relational database reference one another, serving as a foundation for data-integration pipelines, semantic layers, and language-model agents; however, many real-world databases leave them undeclared. Without declared foreign keys, data-integration and ontology-construction pipelines fail to link entities across tables, business-intelligence layers and text-to-SQL agents generate invalid or incomplete joins, and practitioners face costly manual schema reverse-engineering. Solving this problem automatically is challenging because value inclusion alone does not imply reference: auto-increment integer keys induce pervasive coincidental inclusions (across 259 public databases, 80% of declared foreign keys target integer keys, and 64% of these face at least one competing key that also contains the source values), rival tables often share identical key sets under abbreviated or opaque column names, and querying large language models (LLMs) across candidate column pairs incurs high token cost and uncalibrated false discoveries. In this paper, to address these problems, we first formalize these structural traps with two theoretical results: a *hypergeometric chance model* with conservative domain-scale estimators that quantifies the statistical significance of an observed inclusion, and an *identifiability bound* proving that when ν candidate targets are indistinguishable in the data, no data-only method can select the correct target with probability above $1/\nu$—a theoretical ceiling that empirical accuracy without schema names directly reaches. Guided by these results, we propose RefSel, a five-stage pipeline from raw tables to risk-controlled foreign keys that combines recall-first SQL semi-join candidate generation (Stage 1), two-pass context scoring with inclusion significance and out-of-fold relational stacking (Stage 2), exact whole-database CP-SAT joint decoding (Stage 3), competition-aware confidence decomposition with group-level Learn-then-Test risk control that bounds the expected false-discovery proportion (FDP) of retained foreign keys (Stages 4–5), and language-model evidence with bounded influence: a pre-decoding prior that shifts candidate scores by at most a fixed bound and a post-decoding verifier that vetoes implausible edges. Across 355 internal databases and seven external benchmarks, data-only RefSel reaches 0.897 macro F1 with schema names and 0.715–0.762 anonymously, outperforming five published rule-based, learning-based, and holistic baselines (Fast-FK, Rostin, HoPF, MC-FK, and Auto-BI) by $+6.2$ to $+28.0$ F1 points, while surpassing LLM-FK—a recent LLM-based foreign-key discovery method—by $+13.7$ points on anonymized schemas at zero model calls. Across three pre-registered blind rounds on 110 unseen databases, RefSel with bounded LLM evidence outperforms same-model LLM-FK by $+13.1$ F1 points on conventional schemas (0.819 vs. 0.688) and $+9.2$ points on unique-target keys while issuing $7.5\times$ fewer model calls and consuming $< 1/20$ of the input tokens, preserves accuracy when upgrading to a $4\times$ costlier model where LLM-FK degrades, and bounds realized FDP below $\alpha = 0.20$.

Index Terms—Foreign-key discovery, inclusion dependencies, data profiling, schema inference, structured prediction, risk control, large language models

1 Introduction

Foreign keys specify how rows in one table reference rows in another, and most tools that consume relational databases depend on them: data-integration and profiling pipelines align sources along foreign keys [1], business-intelligence tools turn them into join graphs [2], and language-model agents use them to choose join paths [3]. Yet many databases do not declare them. In SchemaPile, a corpus of 221,171 public schemas, 45.9% declare no foreign key at all [4]. Applications often enforce references only in their own code [5], cloud catalogs keep keys as unenforced metadata [6], and exports to flat files drop constraints altogether. Without declared foreign keys, downstream integration and ontology pipelines miss cross-table links, semantic layers require costly manual reverse-engineering, and text-to-SQL agents hallucinate or omit joins; automating recovery must both find the missing references and bound false discoveries before unattended use.

A database instance gives a strong necessary condition: the values of a referencing column must occur in the referenced key. Profiling algorithms find inclusion dependencies (INDs) and unique column combinations (UCCs) efficiently [7], [8], [9]. The difficulty is that *value inclusion does not imply reference*. In MySQL Sakila [10], six tables contain 81 exact unary inclusions into unique columns, and only nine are declared foreign keys. All 599 values of `customer_id` fall inside the 600 values of `city_id`, although customers do not reference cities; `payment.staff_id` takes $\{1, 2\}$ and references `staff.staff_id`, while `store.store_id` holds the same two values. Two mechanisms produce such traps. *Coincidental inclusion*: surrogate keys are auto-increment integers starting at 1, so a column with values in $\{1, \ldots, m\}$ falls inside every such key with at least m rows. Across 259 public databases, 80% of declared unary foreign keys target integer keys, and 64% of these have a competing

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Manuscript draft, October 2026.

key containing all source values (vs. 12% for non-integer keys). *Value-equivalent targets*: tables that extend or partition one another share identical key values, so a reference into one is an equally valid inclusion into all of them; 17% of declared foreign keys in BIRD and 31% in Spider face such a rival. In both cases the wrong candidate is indistinguishable from the right one in the data values.

Existing methods handle these traps only in part. Learned and heuristic methods rank INDs with value and name features [11], [12], [13] or select keys jointly with rules [14], and Auto-BI adds a global schema-graph objective [2]. None measures how strong the inclusion evidence is, keeps confidence low when a rival fits equally well, or bounds how many returned keys are wrong. Systems built on large language models (LLMs) read names and sample values [15], [16], [17]; LLM-FK reports F1 of 0.93–1.00 on five benchmarks with names, dropping to 0.53–0.92 without names while issuing 2,588 calls on MusicBrainz alone [15, Tables 1, 3, 11].

Our key observation is that both traps can be quantified. Whether an inclusion is informative depends on how likely it is by chance, which follows from the sizes and types of the value sets; whether a target can be singled out depends on how many rivals share its observable evidence. With ν indistinguishable rivals, no data-only method beats a uniform guess $1/\nu$, and a calibrated method must show this in its confidence.

Guided by this observation, we build RefSel as five components turning a raw database into foreign keys with confidences (Fig. 1). **Stage 1 (candidate generation)** scans the raw tables, finds minimal UCCs as candidate keys, and runs SQL semi-joins keeping every unary or composite column list whose distinct values are at least 90% contained in a candidate key. **Stage 2 (context scoring)** scores each candidate with *inclusion significance* s_e, the surprisal of the observed overlap under a hypergeometric chance model, and stacked schema features. **Stage 3 (joint decoding)** selects keys and foreign keys for the whole database with an exact CP-SAT solver, allowing cycles and self-references. **Stage 4 (confidence)** splits probability across tied targets, and **Stage 5 (retention)** calibrates a threshold with Learn-then-Test so expected FDP stays below α. With names, an optional LLM module adds a prior shifting a group's scores by at most $B = 6$ and a verifier vetoing implausible decoded edges.

This paper makes three contributions:

- *Measuring chance and target ambiguity.* A hypergeometric chance model with conservative, type-aware domain estimates turns coincidental inclusion into a computable score, and an identifiability bound shows that ν data-indistinguishable targets cannot be separated above $1/\nu$; without names, measured accuracy sits at this limit (Section 3.2).
- *Discovery with false-discovery control and bounded LLM evidence.* RefSel combines recall-first profiling, two-pass scoring, exact whole-database decoding, group-level FDP control, and an LLM prior and verifier with capped influence—the first foreign-key discovery method with a false-discovery guarantee calibrated and tested on independent databases (Section 3).
- *A pre-registered, frozen evaluation.* Beyond 355 internal databases, seven external benchmarks, and 24 registered databases, we freeze the program on 283 development databases, calibrate on unseen databases, and test once on 110 databases in three rounds against LLM-FK under the same model and candidates (Section 5).

The experiments show large gains over published baselines and separate what transfers across repositories from what does not. Without LLM calls, RefSel beats the strongest published baselines (Fast-FK, Rostin, HoPF, MC-FK, Auto-BI) by $+6.2$ to $+28.0$ F1 points on BIRD/Spider (0.897 named, 0.715–0.762 anonymous), $+17.4$ on ReDeLEx (0.745 vs. 0.571), and $+13.8$ to $+18.9$ on seven external benchmarks (0.802 named, 0.721 anonymous), and beats anonymous LLM-FK by $+13.7$ points; a Stage-2 ablation (RefSel-base: Stage 1 + 1-pass GBDT + Stage 3 CP-SAT) drives most of this lead ($+4.9$ to $+17.4$ points), while s_e and stacking add $+1.2$ to $+4.7$ points on BIRD/Spider. On 110 unseen databases across three frozen rounds, bounded LLM evidence lifts both Stage 2 variants ($0.630 \rightarrow 0.819$ on conventional schemas in Round 3), beating LLM-FK on conventional schemas (0.819 vs. 0.688), unique-target keys ($+9.2$ points), and anonymous schemas ($+9.2$ points) at $1/7.5$ of its calls and $< 1/20$ of its tokens while keeping realized FDP below $\alpha = 0.20$.

Section 2 defines the problem, Section 3 presents RefSel, Section 4 reviews related work, Section 5 reports the experiments, and Sections 6 and 7 discuss limitations and conclude. Proofs, implementation details, and the full-length experimental report are in an online appendix at https://github.com/wangnengdejiamao/RefSel-online-appendix.

2 Background and Problem Definition

2.1 Keys, inclusions, and foreign keys

A relational database instance D consists of tables $T_1, \ldots, T_n$. For a table A and an ordered list of its columns $X = (x_1, \ldots, x_p)$, let $\mathrm{val}(A.X)$ denote the set of distinct tuples of non-null values that A takes on X.

Definition 1 (Unique column combination)**.** A column list X is a *unique column combination* (*UCC*) of table A if no two rows of A agree on all columns of X and none of those values is null. It is *minimal* if no proper subset of X is a UCC.

Definition 2 (Inclusion dependency)**.** For column lists X of A and Y of B with $|X| = |Y|$, the *inclusion dependency* (*IND*) $A.X \subseteq B.Y$ holds if $\mathrm{val}(A.X) \subseteq \mathrm{val}(B.Y)$. It holds *approximately* with tolerance $\varepsilon \in [0, 1)$ if at most a fraction ε of the tuples in $\mathrm{val}(A.X)$ lies outside $\mathrm{val}(B.Y)$.

A *foreign key* $A.X \rightarrow B.Y$ is a declared reference from a source column list X of table A to a unique key Y of table B. In a referentially consistent instance, every foreign key implies the IND $A.X \subseteq B.Y$, but an IND into a UCC is only a *candidate* reference, because small integer identifiers, dates, and short codes nest inside larger keys by coincidence. SQL allows foreign keys to reference any unique key [18], and real schemas contain self-references ($A = B$) and cycles.

2.2 Candidates and source groups

A *candidate* $e = (A, X, B, Y)$ pairs a source column list $A.X$ with a candidate key Y of B such that $A.X \subseteq B.Y$ holds exactly or approximately; $\mathcal{E}(D)$ denotes the set of candidates that Stage 1 computes from the raw tables of D (Section 3.1). The *source group* $g(e)$ is the set of candidates

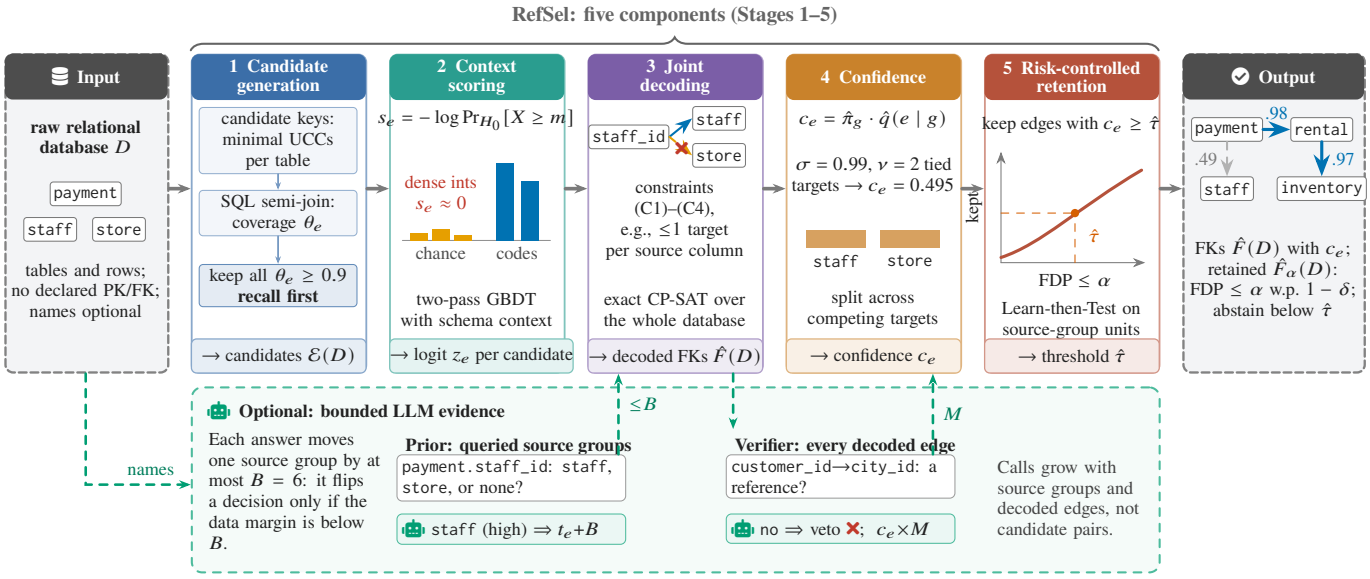


Fig. 1. Overview of RefSel. **Input:** a raw relational database D (tables and rows, no declared primary or foreign keys; names optional). **Five components (Stages 1–5):** (1) candidate generation finds candidate keys (minimal unique column combinations) and keeps every column list whose values are at least 90% contained in one, using SQL semi-joins; (2) context scoring combines inclusion significance s_e with two-pass schema features; (3) joint decoding selects keys and foreign keys for the whole database under constraints (C1)–(C4); (4) confidence c_e splits probability across value-equivalent targets; (5) Learn-then-Test calibrates a retention threshold $\hat{\tau}$. **Output:** foreign keys $\hat{F}(D)$ with confidences and the retained subset $\hat{F}_\alpha(D)$ (confidences are illustrative). **Optional module:** with schema names, an LLM shifts the scores of queried source groups by at most $B = 6$ before Stage 3 and verifies decoded edges before Stages 4–5. The example is the Sakila database.

in $\mathcal{E}(D)$ that share the source table A and source columns X. A source group is the natural decision unit: does the source reference any table, and if so, which competing target is the true one? In Sakila, `payment.staff_id` forms one source group with two candidates, into `staff.staff_id` and `store.store_id`, whose value sets are both $\{1, 2\}$.

2.3 Problem statement

Given an instance D stripped of all declared primary and foreign keys, the task is to output a set of foreign keys $\hat{F}(D) \subseteq \mathcal{E}(D)$, optionally with one selected primary key per table, together with a confidence $c_e \in [0, 1]$ for each predicted edge. When the user specifies a false-discovery tolerance $\alpha \in (0, 1)$, the method also outputs a retained subset $\hat{F}_\alpha(D) \subseteq \hat{F}(D)$ whose expected false-discovery proportion is at most α (Section 3.5). Predictions are compared with the set G_D of foreign keys declared in D.

Naming conditions: In the *original* condition, methods see the table and column names. In the *anonymous* condition, all names are replaced by opaque identifiers (e.g., `t1.c1`) and the column order of every table is permuted, so inference must rely on data values and atomic types alone; this models exported files, obfuscated schemas, and machine-generated tables. Intermediate degradations (prefixes, abbreviations) and distractor tables are evaluated in Online Appendix I.2, and Online Appendix G maps these conditions to operational scenarios.

3 The RefSel Method

RefSel takes a constraint-free database instance D as input and outputs foreign keys $\hat{F}(D)$ with confidences c_e and, for a tolerance α, a retained subset $\hat{F}_\alpha(D)$. Fig. 1 shows the five components between input and output; Algorithm A1 in the online appendix gives the pseudocode. Stage 1 maximizes recall, and Stages 2–5 filter and disambiguate its candidates. Section 3.2 gives the two results that shape Stages 2, 4, and 5.

Notation: For a candidate $e = (A, X, B, Y) \in \mathcal{E}(D)$, let $S_e = \mathrm{val}(A.X)$ with $k_e = |S_e|$, $P_e = \mathrm{val}(B.Y)$ with $N_e = |P_e|$, $m_e = |S_e \cap P_e|$, and coverage $\theta_e = m_e/k_e$. $\mathcal{G}_D$ is the set of source groups and $\mathcal{T}_D$ the set of tables; the subscript e is dropped when clear.

3.1 Recall-first candidate generation (Stage 1)

Stage 1 reads only the raw tables of D and builds $\mathcal{E}(D)$ in two steps; no candidate list is supplied from outside. *Key candidates:* for every table T, we enumerate its minimal UCCs $\mathcal{K}_T$ up to arity three (four in small schemas). Columns that are unique on their non-null values but contain NULLs remain valid targets without entering $\mathcal{K}_T$, and floating-point columns holding only integers are normalized to integers. *Inclusion candidates:* for every source column list $A.X$ and key candidate $B.Y$ of the same arity and compatible types, an SQL semi-join over distinct, type-normalized tuples computes the overlap m_e. Exact containment would miss foreign keys in dirty or partial exports, so we keep

$$\mathcal{E}(D) = \{\, e : \theta_e \geq 1 - \varepsilon \,\} \cup \mathcal{E}^{\mathrm{aux}}(D), \qquad \varepsilon = 0.1, \quad (1)$$

where $\mathcal{E}^{\mathrm{aux}}(D)$ adds two boundary cases that fail the 90% threshold: low-cardinality or skewed sources with $\theta_e \geq 0.5$ that miss at most one distinct value or match at least 90% of the rows, and single-valued sources, flagged as constant. All thresholds were fixed before the experiments. Candidate enumeration runs in descending coverage under size-dependent budgets (Online Appendix B). A foreign key missing from $\mathcal{E}(D)$ cannot be recovered later; Table A2 lists the candidate recall of each collection.

3.2 Theory: coincidental inclusion and identifiability

A chance model for inclusion: When a source column takes values in $\{1, \ldots, 599\}$ and a target key holds $\{1, \ldots, 600\}$, exact inclusion holds because both are dense counters. Let $\mathcal{U}$ be the domain from which S would be drawn if $A.X$ had no relationship with $B.Y$, with size $u = |\mathcal{U}| \geq N$. Under the null hypothesis H_0 of coincidental inclusion, S is a uniformly random k-subset of $\mathcal{U}$, independent of $P \subseteq \mathcal{U}$, so the overlap $X = |S \cap P|$ is hypergeometric:

$$\Pr_{H_0}[X = i] = \binom{N}{i}\binom{u-N}{k-i}\Big/\binom{u}{k}. \tag{2}$$

The *inclusion significance* of e is the surprisal of observing at least m matches by chance:

$$s_e = -\log \Pr_{H_0}[X \geq m]. \tag{3}$$

A large s_e means the overlap is unlikely without a reference; $s_e \approx 0$ means target density N/u explains it (proofs in Online Appendix A).

Lemma 1 (Chance inclusion)**.** *Let $q = N/u$ and $\theta = m/k$. (a) If $m = k$, then $\Pr_{H_0}[X = k] \leq q^k$, so $s_e \geq k \log(u/N)$. (b) If $\theta > q$, then $s_e \geq k\,\mathrm{KL}(\theta \,\|\, q)$, where* KL *is the binary Kullback–Leibler divergence. (c) Under H_0 with the true u, $\Pr_{H_0}[s_e \geq t] \leq e^{-t}$ for $t \geq 0$, so $\mathbb{E}_{H_0}[s_e] \leq 1$.*

A true foreign key over a sparse domain ($u \gg N$) therefore earns significance that grows linearly in k, whereas a coincidental inclusion has expected surprisal at most 1. Statement (c) carries over to an estimate $\hat{u}$ only if $\hat{u} \leq u$, so we take the minimum over estimators. *Domain estimates:* we use the global integer or date span of $S_e \cup P_e$ (rng), the local span of S_e (loc), the number of distinct values of the same type in D (typ), and, for strings, the size of the inferred character pattern (pat), each at least $N + k - m$, and set

$$s_e^{\text{cons}} = \min\{\, s_e^{\text{loc}},\, s_e^{\text{rng}},\, s_e^{\text{typ}},\, s_e^{\text{pat}} \,\}. \tag{4}$$

For two dense counters starting at 1, the local estimate gives $\hat{u} \approx N$ and $s_e^{\text{cons}} \approx 0$. Because real non-references can cluster in ways H_0 does not capture, Stage 2 uses s_e^{cons} and its components as features, with no hard cutoff.

Identifiability: Some target ambiguity cannot be resolved from values at all. In the anonymous condition, let O be the full observation (instance with opaque names and permuted columns). For a source group g with candidates $e_1, \ldots, e_\nu$, let $R \in \{e_1, \ldots, e_\nu, \perp\}$ be its true reference ($\perp$: none) and $\pi_g = \Pr[R \neq \perp \mid O] > 0$.

Definition 3 (Indistinguishable targets)**.** Candidates $e_1, \ldots, e_\nu \in g$ are *indistinguishable* in O if, for every permutation σ of $\{1, \ldots, \nu\}$, some relabeling of tables and columns leaves O invariant and maps each e_i to $e_{\sigma(i)}$.

This happens when ν tables share identical key value sets and interchangeable column distributions. We assume that the law of (O, R) is invariant under relabeling identifiers jointly in O and R.

Proposition 1 (Identifiability bound)**.** *If $e_1, \ldots, e_\nu \in g$ are indistinguishable in O, then $\Pr[R = e_i \mid O] = \pi_g/\nu$ for every i. Hence any decision rule that observes only O, deterministic or randomized independently of R given O, selects the true target with probability at most $1/\nu$ given $R \neq \perp$.*

Corollary 1 (Equal split is calibrated)**.** *Under the conditions of Proposition 1, assigning $c_{e_i} = \pi_g/\nu$ to each indistinguishable candidate is calibrated, and assigning more than π_g/ν to a selected target is overconfident.*

The same bound holds for a classifier whose features are relabeling-invariant and identical across the ν targets (Proposition A1 in the online appendix); stacked features that describe how *other* tables use a target can break the symmetry. We use the result twice. Stage 2 receives the *value-equivalence class size* $\nu_e = |\{f \in g(e) : P_f = P_e\}|$, a necessary condition for indistinguishability, as a feature, and Stage 4 divides confidence among targets that remain tied after scoring.

3.3 Two-pass context scoring (Stage 2)

Stage 2 estimates the probability that each candidate is a declared foreign key from three feature families (full list in Online Appendix H.3). *Edge features* describe e alone: coverage θ_e, s_e^{cons} and the individual domain estimates, k_e/N_e, value moments, and, with original names, lexical similarity of (A, X) and (B, Y). *Group features* compare e with its rivals in $g(e)$: rank by coverage and target size, the log-significance gap to the group leader, and ν_e. *Stacked relational features* summarize a first pass over the whole database: a first classifier f_1 produces logits $z_e^{(1)}$, from which we compute 20 features such as the softmax share of e in its group, the first-pass mass of other groups that point to the same target key (hub in-degree), the number of other references leaving the source table, and the foreign-key density of D. A second classifier f_2 uses all three families to output the logit z_e. This is stacked generalization for collective classification [19], [20]; first-pass logits used for training are computed out-of-fold on group-disjoint splits, so f_2 never sees leaked labels. Both classifiers are histogram gradient-boosted trees with a logistic loss weighted so that every database and every source group contributes equally.

3.4 Joint key and foreign-key decoding (Stage 3)

Thresholding z_e per candidate cannot enforce schema validity: candidates of one source group exclude each other, and all references into a table should agree on its key. Stage 3 selects one key per table and a consistent set of foreign keys for the whole database. Binary variables $x_{T\kappa}$ select a key $\kappa \in \mathcal{K}_T \cup \{\perp\}$ for each table, and y_e selects candidate e. The feasible set $\mathcal{C}(D)$ is

$$\begin{aligned}
&\textstyle\sum_{\kappa \in \mathcal{K}_T \cup \{\perp\}} x_{T\kappa} = 1 && \forall T \in \mathcal{T}_D, && \text{(C1)}\\
&y_e \leq x_{B_e, Y_e} && \forall e \text{ with a key target}, && \text{(C2)}\\
&\textstyle\sum_{e \in h} y_e \leq 1 && \forall h \in \mathcal{G}_D, && \text{(C3)}\\
&y_e + y_{\bar{e}} \leq 1 && \forall e,\ \bar{e} \text{ its reverse}. && \text{(C4)}
\end{aligned}$$

(C1) selects at most one key per table, (C2) makes a foreign key into $\mathcal{K}_{B_e}$ point to the selected key, (C3) keeps at most one target per source group, and (C4) forbids two columns referencing each other. No acyclicity is imposed, so cycles and self-references remain feasible. With key scores $a_{T\kappa}$ from a key classifier ($a_{T\perp} = 0$) and shifted logits $t_e = z_e + b$ (b tuned on training folds), Stage 3 solves

$$\max_{(x,y) \in \mathcal{C}(D)} \sum_{T,\kappa} a_{T\kappa} x_{T\kappa} + \sum_e t_e\, y_e \tag{5}$$

with CP-SAT [21] under a 20 s limit. The median solve time is about 1 ms, and all internal databases but one are solved to optimality. The selected edges form $\hat{F}(D)$. Because stacking already feeds hub and competition information into z_e, the objective needs no higher-order terms; learned higher-order potentials did not help (Section 5.4).

3.5 Confidence and risk-controlled retention (Stages 4 and 5)

The logit t_e compares e with selecting no reference, so $\sigma(t_e)$ ignores rivals in $g(e)$. For retention we need the probability that e is the true target among its rivals.

Confidence (Stage 4): For a decoded edge e with group $g = g(e)$, we factor

$$c_e = \hat{\pi}_g \cdot \hat{q}(e \mid g), \qquad \hat{q}(e \mid g) = \frac{\exp(t_e)}{\sum_{f \in g} \exp(t_f)}, \tag{6}$$

where $\hat{\pi}_g = \psi(\max_{f \in g} t_f)$ is an isotonic calibration map fitted on out-of-fold training predictions. When ν candidates tie at the top of g, each has $c_e \le \hat{\pi}_g/\nu$, with equality if the group contains only them. For `payment.staff_id`, a classifier that is 99% sure the column is a reference ($\hat{\pi}_g = 0.99$) but cannot tell `staff` from `store` gives both $\sigma(t_e) \approx 0.99$, whereas Eq. (6) gives $c_e = 0.495$. Section 5.5 measures the calibration of $\hat{\pi}_g$; this does not establish calibration of the product c_e, and Stage 5 does not require it.

Retention (Stage 5): For a threshold τ, Stage 5 keeps $A_\tau(D) = \{e \in \hat{F}(D) : c_e \ge \tau\}$, whose false-discovery proportion is

$$\mathrm{FDP}_\tau(D) = \frac{|A_\tau(D) \setminus G_D|}{\max(1, |A_\tau(D)|)}. \tag{7}$$

Databases from one family or template are dependent, so we group them into *risk units* U_j [22] with loss $L_j(\tau) = |U_j|^{-1} \sum_{D \in U_j} \mathrm{FDP}_\tau(D)$ and control $R(\tau) = \mathbb{E}[L_j(\tau)] \le \alpha$. Given n calibration units and a descending threshold grid, we compute a Hoeffding–Bentkus p-value [23] for $H_\tau : R(\tau) > \alpha$ at each grid point and, following Learn-then-Test (LTT) [24], walk down the grid and select the smallest $\hat{\tau}$ such that $H_{\tau'}$ is rejected at level δ for every $\tau' \ge \hat{\tau}$ (formulas in Online Appendix C). If the strictest point is not rejected, Stage 5 abstains.

Proposition 2 (Risk-controlled retention [24]). *If the n calibration units and a new unit are i.i.d. draws, and the predictor, confidence, and grid are fixed before the calibration labels are read, then with probability at least $1 - \delta$, $R(\hat{\tau}) = \mathbb{E}[L_{\mathrm{new}}(\hat{\tau})] \le \alpha$.*

The bound concerns the expected FDP over new units, not the FDP of every single database. It needs a *frozen program*: Stage 1 rules, f_1, f_2, b, the key scorer, ψ, and the grid must be locked before calibration, which internal cross-validation only approximates; Section 5.7 runs the frozen protocol. The guarantee holds for any fixed score; which score retains more true edges is an empirical question (Section 5.5).

3.6 Bounded language-model evidence

With original names, an LLM knows, for example, that `cust_id` abbreviates `customer.id`. Unconstrained LLM agents, however, can propose joins that contradict the values, and asking about every candidate pair is expensive. RefSel therefore uses an LLM in two capped roles (bottom of Fig. 1); without names, it is not used.

Bounded prior: Let $t_g^{(1)} \ge t_g^{(2)}$ be the two largest shifted logits of group g. The prior queries the groups whose decision margin is at most μ,

$$\mathcal{Q}(D) = \{\, g \in \mathcal{G}_D : \min(|t_g^{(1)}|,\ t_g^{(1)} - t_g^{(2)}) \le \mu \,\}. \tag{8}$$

For each queried group, the LLM sees the schemas and sample values of the source and its candidate targets and names one target or `none` with confidence `high`, `medium`, or `low`. This sets a weight $\varpi_e = \pm\kappa$, $\kappa \in \{1, \frac{1}{2}, \frac{1}{4}\}$, on the named target (or on all candidates for `none`), and Stage 3 uses

$$t_e^{\mathrm{L}} = t_e + B\,\varpi_e, \qquad e \in g \in \mathcal{Q}(D). \tag{9}$$

The margin $\mu = 5$ and the cap $B = 6$ were chosen once by nested cross-validation on 40 development databases; the frozen independent rounds query every group ($\mu = \infty$). Groups are batched, so calls grow with the number of queried groups, not with candidate pairs.

Remark 1 (Bounded influence)**.** Because (C3) lets a feasible solution select at most one candidate per group, the prior changes the objective of any feasible solution by at most $B|\mathcal{Q}(D)|$. The decoded solution is therefore within $2B|\mathcal{Q}(D)|$ of the data optimum, and for a group that shares no contested (C2) constraint, the prior can change the decision only if the data margin is below B.

Verifier: After Stage 3, the LLM reviews each decoded edge in batches, given both schemas, five to ten sample rows, and the Stage 1 statistics (θ_e, k_e, N_e), and answers `yes`, `unsure`, or `no`. A `no` removes the edge (veto), and retention multiplies c_e by $M = 1$, $\frac{1}{2}$, or 0.05 accordingly. On the 284 development databases, 100 of the 109 vetoed edges were false positives. Verifier calls grow with $|\hat{F}(D)|$.

4 Related Work

Dependency profiling and reverse engineering: Data profiling discovers UCCs, functional dependencies, and INDs from instances [1]. SPIDER [7] and BINDER [8] enumerate unary and n-ary INDs at scale, Dürsch et al. [25] benchmark thirteen IND algorithms, DUCC finds minimal UCCs [9], and approximate IND discovery tolerates dangling values [26]. These algorithms decide whether containment holds and emit far more INDs than foreign keys. For functional dependencies, Mandros et al. [27] and Berti-Équille et al. [28] ask when a dependency exceeds chance; RefSel brings this question to inclusion with a hypergeometric null model. Early reverse-engineering systems recovered keys from instances, code, and query logs with a human in the loop [29], [30].

Learned and heuristic foreign-key discovery: Rostin et al. [11] train classifiers over unary INDs with value and name features and note that dense counters starting at 1 cause many false inclusions. MC-FK [12] tests whether source values look like a uniform sample *from the target key*; a prefix range $\{1, \ldots, 599\}$ passes this test against $\{1, \ldots, 600\}$, which our domain-based null model scores near zero. Fast-FK [13] ranks unary candidates with lightweight heuristics,

HoPF [14] infers primary and foreign keys jointly with holistic rules, and Auto-BI [2] calibrates local join probabilities and extracts an acyclic schema graph with a k-MCA objective. RefSel shares the joint-key view of HoPF and the local-then-global design of Auto-BI, and adds inclusion significance with relational stacking, an exact decoder that allows cycles and self-references, and confidence with finite-sample risk control.

LLM-based discovery: LLM-FK [15] coordinates profiling, interpretation, refinement, and verification agents; Chorus [16] and DBAutoDoc [31] use foundation models for join discovery and schema documentation; CGM [17] restricts the LLM to a constraint-feasible set; and Bodensohn et al. [32] show that LLM accuracy drops on noisy enterprise schemas. These systems depend on informative names and spend calls per candidate pair or agent turn. RefSel works without names and, with names, calls the LLM only for capped priors on source groups and checks of decoded edges. Schema matching and join discovery [33], [34], [35], [36] find joinable column pairs across datasets and do not decide directed references within one schema.

Risk control and structured prediction: Conformal prediction gives distribution-free guarantees under exchangeability [37], [38]; RCPS [23] and Learn-then-Test [24] extend finite-sample control to general losses by testing over a threshold grid, and conformal selection controls false discoveries when screening units [39], [40]. Candidate edges within one schema are dependent, so, following hierarchical exchangeability [22], we define risk units at the level of database families. Stage 2 follows stacked generalization for collective classification [19], [20], and Stage 3 is an exact integer program in the spirit of structured prediction [41].

5 Experiments

We ask five questions: effectiveness on shared candidates (**RQ1**, Section 5.3), mechanism (**RQ2**, Section 5.4), risk control under cross-validation (**RQ3**, Section 5.5), generalization and cost (**RQ4**, Section 5.6), and transfer of the frozen pipeline to new repositories (**RQ5**, Section 5.7). Online Appendix I gives the full-length report.

5.1 Metrics

Published studies compute precision, recall, and F1 against different ground-truth denominators, which changes the reported numbers substantially. Our primary *whole-database* protocol compares predicted foreign keys with all declared foreign keys G_D, so a declared key that never enters $\mathcal{E}(D)$, for instance because its source table is empty, counts as a false negative. The *evaluable-key* protocol keeps only declared keys whose source and target columns are non-empty, as in HoPF [14], and the *candidate-domain* protocol keeps only declared keys in $\mathcal{E}(D)$, as LLM-FK does [15]. *Macro F1* averages per-database F1 over databases with $G_D \neq \emptyset$; *pooled precision* and *pooled recall* sum counts over all databases, including those without foreign keys, so false alarms there are penalized. For retention we report correct foreign keys retained per database and the realized FDP (Eq. (7)) averaged over risk units.

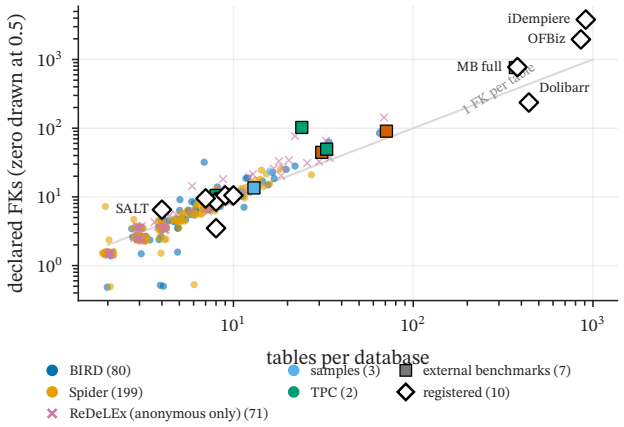


Fig. 2. Benchmark landscape: databases, tables, and declared foreign keys per family, including the external and registered sets.

5.2 Setup

Data: The internal collection has 355 non-empty databases: 80 from BIRD [42], 199 from Spider [43], 71 from ReDeLEx [44], [45], three sample databases such as Sakila, and two TPC benchmarks. Copies and variants of one schema form a source group, and each group is assigned whole to one of five outer folds. All 355 databases are evaluated anonymously and the 284 outside ReDeLEx also with original names. ReDeLEx stores re-indexed dense surrogate keys and is reported separately; internal conclusions rest on the 279 BIRD and Spider databases. The external set has the five benchmarks of LLM-FK (TPC-H, Northwind, TPC-E, AdventureWorks OLTP, MusicBrainz) plus AdventureWorks DW and TPC-DS; these were inspected during development. The registered set has ten databases whose data, labels, and model configuration were hashed before a single run: five public schemas, three open-source ERP systems (Dolibarr, Apache OFBiz, iDempiere), SAP SALT, and the full MusicBrainz dump, plus 14 CTU databases [46] registered later. Section 5.7 describes the independent rounds. Fig. 2 summarizes all collections, and Table A2 lists them in detail.

Baselines and Stage-2 ablation: All methods share Stage 1 candidates unless noted, isolating scoring and decoding. We compare RefSel against seven external baselines: an *inclusion rule* (highest-coverage target per source group); MC-FK [12] and Fast-FK [13], tuned on training folds; a *Rostin-style* classifier [11]; HoPF's holistic rules [14]; *Auto-BI-lite*, feeding our local scores into Auto-BI's acyclic k-MCA-CC objective [2]; and, on external and independent data, LLM-FK [15] with its published prompts and the same backbone model. To isolate Stage 2's s_e and stacking (and Stage 4's c_e), we also track an internal Stage-2 ablation, *RefSel-base* (labeled `GBDT` in figures), which keeps our Stage 1 candidates and Stage 3 CP-SAT decoder (C1)–(C4) with a single-pass GBDT scorer without s_e, stacking, or c_e (tuned on a larger grid).

Protocol and implementation: Paired differences use two-sided Wilcoxon signed-rank tests over duplicate-group means with Holm correction within each block, and 95% bootstrap intervals over 2,000 group resamples; two further fold partitions measure partition variance (Table A3). Under our pre-specified admission rule, s_e and stacking enter the *rule-selected* configuration if their inner-validation gain over

RefSel-base has a 95% interval above zero. This held with names (where rule-selected and *full* coincide) but not without names (where rule-selected falls back to RefSel-base). Because full scored higher on the anonymous test folds (0.762 vs. 0.715), we locked it as a second configuration before any external, registered, or independent run; its anonymous internal numbers are post hoc (‡), and all later runs are confirmatory. RefSel uses Python 3.12, DuckDB 1.5, scikit-learn 1.9, OR-Tools CP-SAT 9.15 [21], and DeepSeek V4.1 Flash (temperature 0, prompt caching) on an Apple M2 (8 cores, 24 GB).

5.3 RQ1: Effectiveness and Stage-2 Ablation

Across all three internal columns (Table 1), RefSel beats every published baseline by a wide margin. With original names on BIRD and Spider, RefSel reaches macro F1 0.897, ahead of Fast-FK (0.827) by $+0.070$ overall ($+0.062$ paired per group, CI $[0.036, 0.088]$, $p < 10^{-4}$), Rostin-style (0.806) by $+0.091$, Auto-BI-lite (0.769) by $+0.128$, HoPF-style (0.759) by $+0.138$, and MC-FK (0.617) by $+0.280$. Without names, lexical baselines collapse (Rostin-style 0.543, Fast-FK 0.649, HoPF-style 0.555) and MC-FK stays at 0.617, whereas RefSel reaches 0.715 (rule-selected) and 0.762 (full), beating Fast-FK by $+0.066$ to $+0.113$ and Auto-BI-lite by $+0.082$ to $+0.129$. On ReDeLEx, RefSel reaches 0.745–0.746, beating Rostin-style (0.571) and Fast-FK (0.570) by $+0.174$. Within our architecture, the single-pass ablation RefSel-base (0.876, 0.715, 0.745) already beats all published baselines by $+0.049$ to $+0.174$, while s_e and stacking add $+0.021$ with names on P0 (CI $[0.008, 0.035]$, $p = 0.004$; $+0.012$ across partitions, CI $[0.002, 0.024]$) and $+0.047$ anonymously on P0 ($+0.024$ across partitions, CI $[0.007, 0.045]$).

Fig. 3 adds two stress tests. Abbreviated names cost about three points (0.865); most of the drop comes only when names disappear. Synthetic distractor tables, whose integer columns nest inside existing keys, lower F1 to 0.782 with names and 0.544 without and raise the realized FDP at $\alpha = 0.20$ from 0.12 to 0.27 and from 0.13 to 0.38, a shift that retention calibrated on clean data cannot absorb.

5.4 RQ2: Mechanism

Where errors come from: Table 2 splits BIRD and Spider predictions into true positives, wrong targets (rival target chosen), missed references, and spurious references. Stage 1 misses 70 of 1,520 declared keys (4.6%), and with an oracle scorer (C1)–(C4) keep all but 17 of the 2,523 gold keys in $\mathcal{E}(D)$. Compared with Fast-FK, RefSel recovers 153 more true keys with names (1,312 vs. 1,159, cutting misses $316 \to 151$) and 224 more anonymously (1,096 vs. 872) while cutting spurious edges ($248 \to 196$ named, $432 \to 279$ anonymous). Against RefSel-base, s_e and stacking cut spurious edges ($222 \to 196$) and (C2) key losses ($33 \to 20$) with names and recover 72 more true keys without names ($1{,}024 \to 1{,}096$). Without names, wrong targets jump from 57 to 230, and 177 of them (77%) fall between targets with identical value sets.

The identifiability limit: Fig. 4 (right) groups decoded source groups by the number ν of value-equivalent targets. Without names, the share with the correct target is 0.95, 0.59, 0.46, and 0.23 for $\nu = 1, 2, 3, \geq 4$, close to $1/\nu = 1.00$, 0.50, 0.33, and 0.18, and the 95% interval contains $1/\nu$ for every $\nu \geq 2$. With names, the shares are 0.98, 0.95, 0.66, and 0.99. Routing all references of an equivalence class to one hub table does not help: even an oracle hub gains only 1.8 F1 points, because in over a third of multi-target classes different keys reference different tables. Stage 4 therefore expresses unresolved competition in c_e, and Stage 3 has no hub constraint.

Inclusion significance: Raw coverage θ_e barely separates true from false candidates (AUC 0.52 on BIRD). Fig. 4 (left) shows s_e^{cons} behaving as Lemma 1 predicts. On sparse integer keys it separates the two with AUC 0.83; on dense counters it gives $s_e < 1$ to 87% of true and 98% of false candidates, marking such inclusions as uninformative either way; on string keys it does not separate them (AUC 0.46), because false string inclusions come from shared vocabularies such as country codes, a form of value equivalence. The ablations (Table A4) attribute most of the Stage-2 gain over RefSel-base to relational stacking. Learned higher-order decoder potentials, an unsupervised prior-shift correction [47], hub-consistent decoding (-0.039), and inbound features for the key classifier brought no gain and were left out.

5.5 RQ3: Risk control under cross-validation

Here ψ and $\hat{\tau}$ are fitted on inner folds, an approximation of the frozen protocol (Online Appendix C). On the BIRD and Spider test folds, realized FDP stays below α at every level (Fig. 5): 0.112 with names and 0.126 without at $\alpha = 0.20$, and 0.04 and 0.03 at $\alpha = 0.10$, while 4.70 and 2.76 correct foreign keys per database are retained at $\alpha = 0.20$. The bound concerns the mean over units: 57 of the 279 databases exceed 0.20 on their own. Away from the calibration distribution the control weakens. Held-out families stay controlled except named BIRD (0.22), the seven external benchmarks stay at 0.16 and 0.07, and the ten registered databases reach 0.36, driven by the sparse ERP demonstration schemas (0.59 and 0.63).

c_e and $\sigma(t_e)$ rank the same decoded edges differently (Fig. 6), and neither dominates: over 20 paired settings, c_e retains more true edges in 7, $\sigma(t_e)$ in 5, and 8 tie. At $\alpha = 0.10$, c_e retains slightly fewer (-0.26 per database with names, -0.43 without), because tied groups receive $c_e \leq 0.5$ and fall below strict thresholds even when the decoder chose correctly. A better Stage 2 scorer matters more than the confidence formula: without names, RefSel (full) retains 3.14 keys per database at $\alpha = 0.20$, against 2.76 for the rule-selected configuration. The isotonic estimate $\hat{\pi}_g$ has an expected calibration error between 0.02 and 0.11 across families.

5.6 RQ4: Generalization and cost

External benchmarks and LLM-FK: Without an LLM, RefSel reaches macro F1 0.802 with names and 0.703 without (0.721 full) on the seven external benchmarks (RefSel-base: 0.767, 0.703), beating Fast-FK (0.664, 0.532) by $+0.138$ to $+0.189$ and MC-FK (0.478) by $+0.324$ (Table A5). Fig. 7 compares RefSel with LLM-FK on LLM-FK's five benchmarks. Without names, data-only RefSel beats our same-model rerun of LLM-FK by $+0.137$ (0.711 vs. 0.574); on the 370-table MusicBrainz, LLM-FK spends 9,790 calls to reach 0.162 (vs.

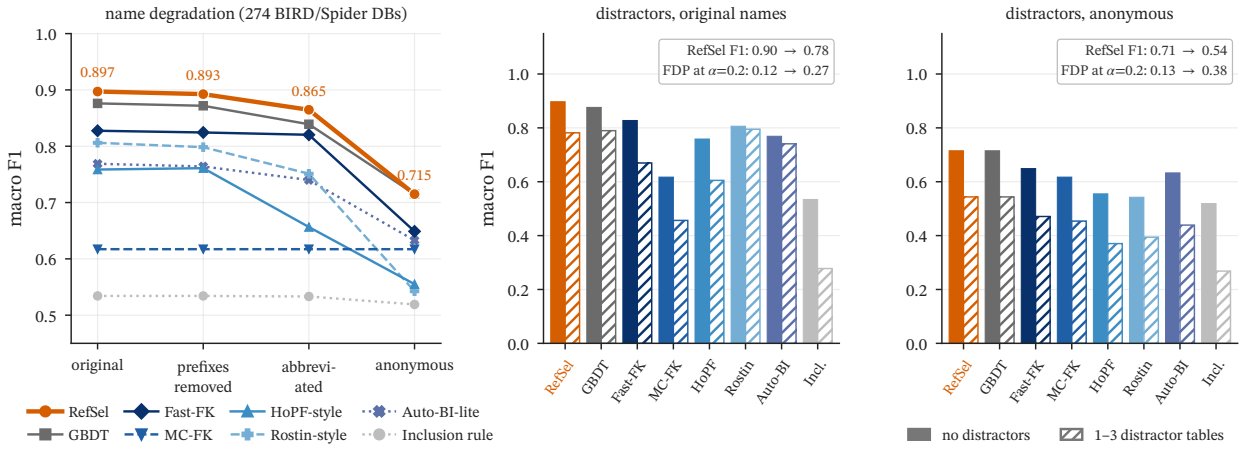


Fig. 3. Left: macro F1 of RefSel, its Stage-2 ablation (GBDT: RefSel-base), and every external baseline of Table 1 as names degrade from original to anonymous (274 BIRD and Spider databases; outer-fold models). Right: the same databases without (solid) and with (hatched) one to three synthetic distractor tables whose integer columns fall inside existing key ranges; boxes give RefSel's F1 and realized FDP at $\alpha = 0.2$.

TABLE 1
Macro F1 and pooled precision/recall on shared candidates (whole-database protocol, five outer folds); best pre-specified F1 in bold. Top block: external baselines; bottom block: RefSel and its Stage-2 ablation (RefSel-base: Stage 1 + 1-pass GBDT + Stage 3 CP-SAT). With names the two RefSel configurations coincide; without names rule-selected equals RefSel-base, and full was fixed after these folds were seen (‡).

	BIRD/Spider, original			BIRD/Spider, anonymous			ReDeLEx, anonymous		
Method	F1	P	R	F1	P	R	F1	P	R
Inclusion rule	0.534	0.134	0.889	0.519	0.132	0.874	0.158	0.020	0.933
MC-FK [12]	0.617	0.304	0.841	0.617	0.304	0.841	0.465	0.050	0.867
Fast-FK [13]	0.827	0.799	0.762	0.649	0.552	0.574	0.570	0.244	0.406
Rostin-style [11]	0.806	0.893	0.718	0.543	0.790	0.389	0.571	0.757	0.212
HoPF-style [14]	0.759	0.578	0.706	0.555	0.418	0.543	0.438	0.151	0.311
Auto-BI-lite [2]	0.769	0.871	0.552	0.633	0.707	0.446	0.529	0.523	0.221
RefSel-base (1-pass GBDT ablation)	0.876	0.821	0.855	**0.715**	0.662	0.674	**0.745**	0.518	0.613
RefSel (ours, rule-selected)	**0.897**	0.838	0.863	**0.715**	0.662	0.674	**0.745**	0.518	0.613
RefSel (ours, full: $+s_e+$ stacking)	**0.897**	0.838	0.863	0.762‡	0.686	0.721	0.746‡	0.536	0.628
Gain of RefSel over best baseline	*+0.070*	*+0.039*	*+0.101*	*+0.066/+0.113‡*	*+0.110*	*+0.100*	*+0.174/+0.175‡*	–	*+0.207*

TABLE 2
Errors on BIRD and Spider (1,520 declared foreign keys; 1,518 in the anonymous condition, where two declared keys name columns that do not exist). Missed references are split into no candidate / excluded by (C2) / scored out; a source with a wrong prediction counts as a wrong target even if the gold target has no candidate.

Names	Method	TP	Wrong tgt	Missed	Spurious
original	Fast-FK (baseline)	1159	45	316 (66/0/250)	248
original	RefSel-base (1-pass GBDT)	1299	61	160 (66/33/61)	222
original	RefSel	1312	57	151 (66/20/65)	196
anonymous	Fast-FK (baseline)	872	277	369 (64/0/305)	432
anonymous	RefSel (rule-sel. / base)	1024	230	264 (65/37/162)	293
anonymous	RefSel (full)	1096	222	200 (65/19/116)	279

0.428 whole-database and 0.575 evaluable-key F1 for data-only RefSel). With names, LLM-FK reaches 0.877 and data-only RefSel 0.790; the selective prior ($\mu = 5$) brings RefSel to 0.872 with 846 calls against 13,505 (16× fewer; Fig. 8), and full prior+veto reaches 0.892 (beating LLM-FK's 0.877 at 1/13 of the calls). The prior hurts on AdventureWorks DW (0.755 → 0.500), where the LLM prefers a readable date column over the integer surrogate key. With the 4× costlier DeepSeek V4 Pro [48], RefSel+LLM barely moves (0.892 → 0.880), while LLM-FK drops from 0.877 to 0.828 with names and from 0.574 to 0.477 without (Fig. 9).

Registered databases: On the ten registered databases (Table A8), RefSel reaches 0.61 (0.60 for RefSel-base, vs. 0.59 for Rostin-style and Fast-FK), scoring 0.76 on the five public schemas, 0.786 evaluable-key F1 on the 857-table Apache OFBiz, and 1.000 on the two SAP SALT keys with unique targets in the extract. It fails on sparse ERP demonstration installs: in Dolibarr, most tables hold 1–3 seed rows, making nearly every column a UCC that low-cardinality columns elsewhere fall into (557 of 659 false positives come from constant or binary sources). On the 14 registered CTU databases (Table A9), both RefSel (0.765 named, 0.700 anonymous) and RefSel-base (0.810, 0.700; $p = 0.31$) beat every published baseline: Fast-FK (0.723, 0.654), MC-FK (0.689), Rostin-style (0.624, 0.614), HoPF-style (0.581, 0.616), and Auto-BI-lite (0.571, 0.575).

Downstream use and cost: Adding RefSel's foreign keys to the schema prompt of DeepSeek V4.1 Flash on the 1,070 BIRD join questions raises execution accuracy from 0.539 without keys to 0.545 (vs. 0.550 with declared keys; not significant, since BIRD's small, well-named schemas let the model infer most joins). Stage 1 semi-joins dominate runtime: Stage 3 takes ∼1 ms and Stage 2 <1 s, whereas Stage 1 takes <1 s on typical schemas, 19 min on Dolibarr

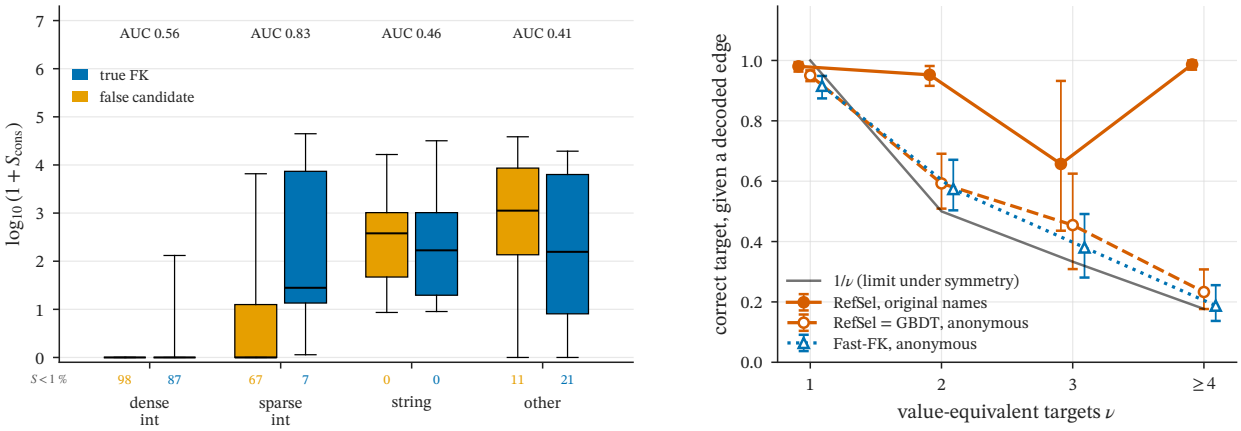


Fig. 4. Coincidental inclusion and ambiguity on BIRD and Spider. Left: inclusion significance of true foreign keys and false candidates by key type, with the AUC of each type and the share of candidates with $s_e < 1$. Right: for source groups with a decoded edge, the share whose decoded target is the declared one, by the number ν of value-equivalent targets, with 95% intervals; without names every method follows the limit $1/\nu$ of Proposition 1.

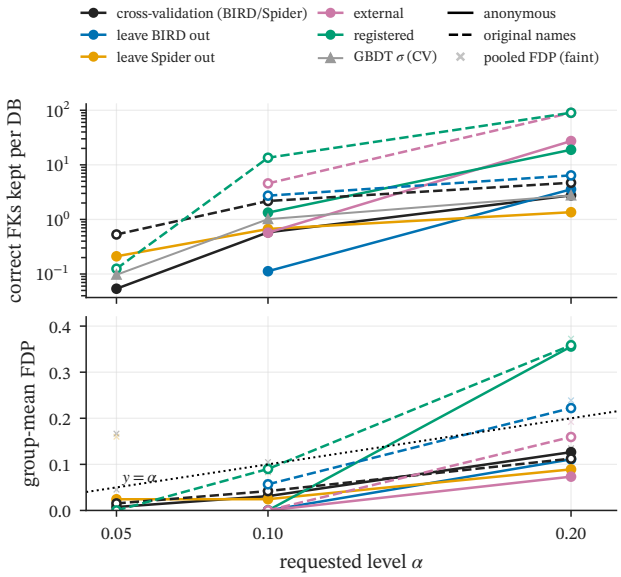


Fig. 5. Correct foreign keys retained per database (top) and realized false-discovery proportion (bottom) against the requested level α, on test folds, held-out families, and external databases. Levels at which nothing is retained are omitted.

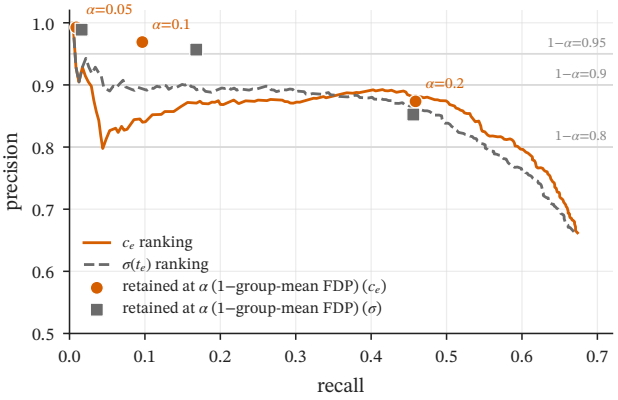


Fig. 6. Precision–recall curves in the anonymous condition. Markers show the operating points retained at $\alpha \in \{0.05, 0.1, 0.2\}$ with c_e and with $\sigma(t_e)$.

(5,755 columns), and hits the 2-hour cap on iDempiere and TPC-H/TPC-DS SF10 (Fig. 10). Running the LLM prior on all seven external benchmarks costs about $1.

5.7 RQ5: Frozen out-of-repository evaluation

Protocol: We split databases by repository origin into three disjoint parts. The *development* part (**D**) is the 283 internal databases outside ReDeLEx; every learned component and the threshold grid are trained on it, and hashes of all model files and code are sealed before any calibration or test label is read. The frozen program runs once on a *calibration* part (**C**), where LTT selects $\hat{\tau}$ at $\delta = 0.10$, and once on a *test* part (**T**); predictions are sealed before labels are read, and every metric is recomputed independently from the sealed files. Rounds 1 and 2 calibrate on 29 CTU databases (24 risk units). Round 1 tests on nine databases from RelBench [49] and 4DBInfer [50]. Round 2 tests on 41 databases in ten source groups (Spider 2.0, WikiDBs [51],

TABLE 3
Frozen rounds, test part: decoded macro F1 over all test databases for RefSel (Full) and its single-pass Stage-2 ablation (RefSel-base, labeled `GBDT` in Fig. 11), against the external baseline LLM-FK, plus the largest realized FDP among retention lines at $\alpha = 0.20/0.10$. +LLM: bounded prior and verifier (original names only). LLM-FK counts databases it may not process as empty. *Post hoc.

		RefSel (data)		RefSel (+LLM)		Baseline	
Round (C/T)	Names	Base	Full	Base	Full	LLM-FK	FDP
1 (29/9)	original	0.952*	0.966*	–	–	–	0.00 / 0.00
	anonymous	0.878*	0.905*	–	–	–	
2 (29/41)	original	0.456	0.438	0.465*	**0.493**	0.411	0.12 / 0.12
	anonymous	0.400	0.391	–	–	0.310	
3 (89/60)	original	0.370	0.402	0.465	**0.476**	0.430	0.16 / 0.09
	anonymous	0.352	0.362	–	–	0.270	

MIMIC-IV demo, 4DBInfer, five tutorial schemas) and adds the LLM prior ($\mu = \infty$, $B = 6$) and verifier. Round 3 draws 149 new databases (32 conventional schemas and 117 WikiDBs), splits them by a registered random split into 89 calibration and 60 test units, and registers four endpoints: **P1**, realized FDP of every retention line at $\alpha \in \{0.10, 0.20\}$; **P2**, correct keys retained at $\alpha = 0.20$ by RefSel-base with prior and verifier; **P3**, the F1 gain of the LLM evidence for each Stage 2 scorer (RefSel and RefSel-base); and **P4**, RefSel

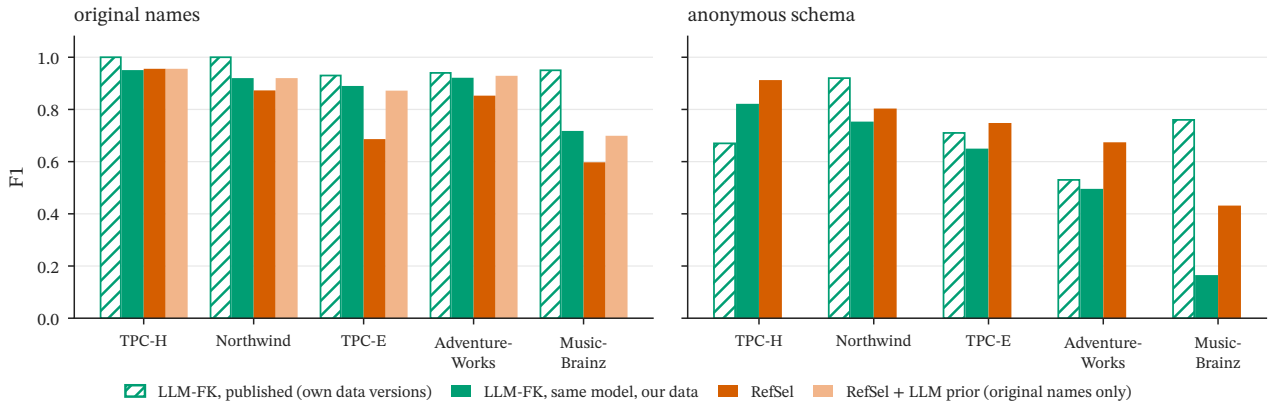


Fig. 7. Comparison with LLM-FK on its five benchmarks, with original names (left) and anonymized schemas (right). Hatched bars are the values published for LLM-FK on its own data versions [15]; solid bars are runs on our data versions under the whole-database protocol. The LLM prior is used with original names only.

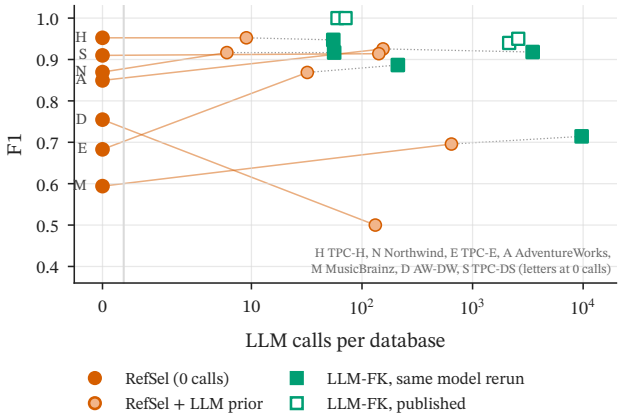


Fig. 8. F1 against the number of LLM calls per database (log scale) for RefSel, RefSel with the LLM prior, the LLM-FK rerun with the same model, and the call counts published for LLM-FK.

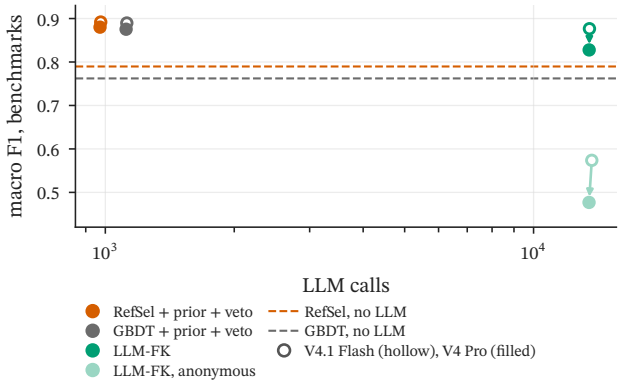


Fig. 9. Choice of model: macro F1 with original names on LLM-FK's five benchmarks against LLM calls (log scale). Arrows lead from V4.1 Flash (hollow) to V4 Pro (filled); lighter arrow: anonymous LLM-FK. Dashed lines: RefSel and its Stage-2 ablation (GBDT: RefSel-base) without LLM calls (prior+veto lines share calls).

with prior and verifier against the external baseline LLM-FK. Databases whose licenses forbid external calls receive no LLM call. Table 3 summarizes the three rounds; Online Appendix I.6 reports each in full.

Risk (P1): At $\alpha = 0.20$, realized FDP on the test part stays below target for every reported line in all three rounds:

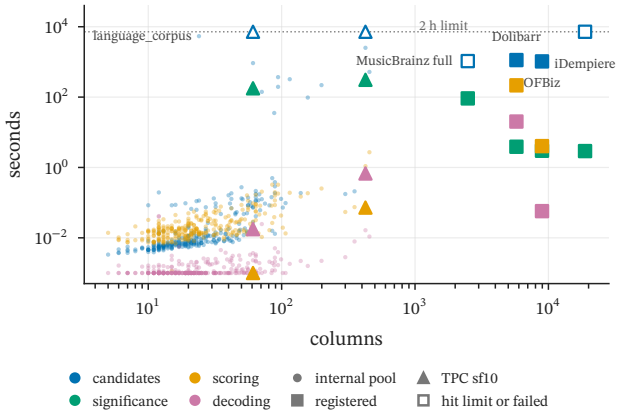


Fig. 10. Runtime of each stage against the number of columns (log–log). Hollow markers: runs that reached the time or memory budget.

0.00 in Round 1, at most 0.12 in Round 2, and at most 0.16 in Round 3, where one-sided 90% bootstrap upper bounds reach 0.20. The one exceedance is at the secondary level $\alpha = 0.10$, for the anonymous $\sigma(t_e)$ line of Round 2 (0.122).

Retention: In Round 1, thresholding RefSel with c_e retained 1.89 correct keys per database while RefSel-base with $\sigma(t_e)$ retained 6.33 ($p = 0.03$). With 24 calibration units, the test at $\alpha = 0.20$ tolerates an empirical FDP of only 0.049, so two confident errors on small CTU schemas stop the threshold walk; our audit found that six of the ten confident "errors" are genuine undeclared references (Online Appendix D). In Round 3, P2 names RefSel-base with prior and verifier, but the saved retention report thresholds RefSel-base edges decoded with the verifier and no prior (0.58 keys per database under either calibration set), so P2 as registered is not established. A supplementary contrast on the same test predictions (Fig. 11(b)) shows the effect of a larger pool: verifier-weighted $\sigma(t_e)$ for RefSel retains 0.55 keys per database when calibrated on CTU vs. 1.97 on the 89 Round 3 units (3.6× more, FDP 0.14), mixing calibration size and source composition.

Bounded LLM evidence (P3) and Stage-2 ablation: In Round 1, data-only RefSel reaches 0.966 named and 0.905 anonymous (+0.014 and +0.027 over RefSel-base, +0.124 on CTU-29). In Rounds 2–3, the bounded prior and verifier

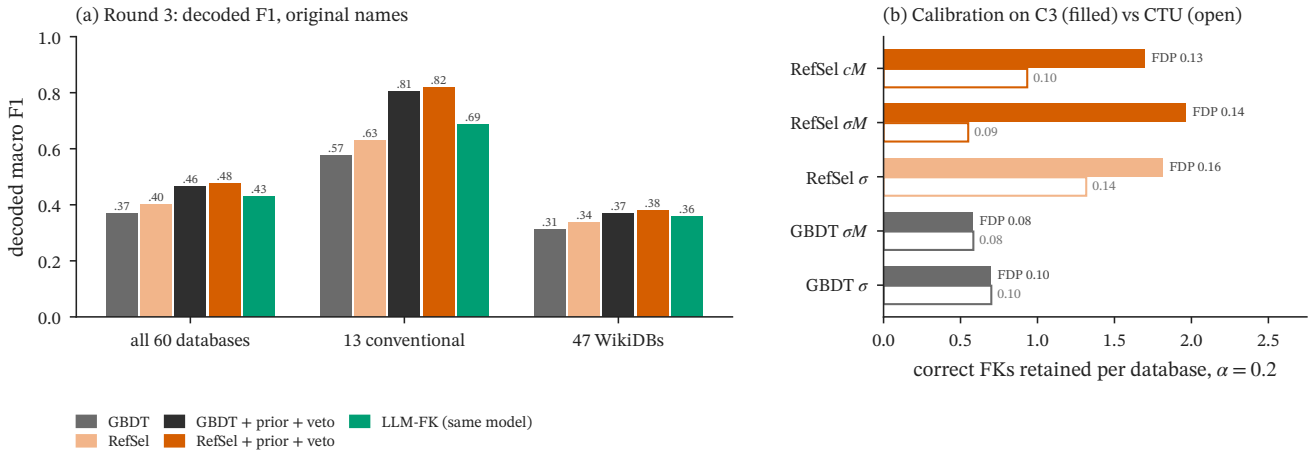


Fig. 11. Third independent round, test part. (a) Decoded macro F1 with original names over all 60 databases, the 13 conventional ones, and the 47 WikiDBs databases for RefSel, its Stage-2 ablation (`GBDT`: RefSel-base), and the baseline LLM-FK (shared Stage 1 candidates and LLM responses). (b) Cross-calibration contrasts: correct keys retained per database at $\alpha = 0.2$ calibrated on Round 3 (filled, 89 units) or CTU (open, 24 units); labels give realized FDP. M: verifier weight (no prior on `GBDT` σM).

lift both Stage 2 variants: in Round 2, RefSel+LLM rises from 0.438 to 0.493 ($+0.037$ over data-only RefSel-base, $p = 0.016$; 0.897 on 11 conventional schemas); in Round 3 (Fig. 11(a)), RefSel+LLM rises from 0.402 to 0.476 (CI $[0.025, 0.121]$) and RefSel-base+LLM from 0.370 to 0.465 (both Holm $p < 10^{-4}$), gaining $+0.19$ ($0.630 \to 0.819$) and $+0.23$ ($0.575 \to 0.806$) on the 13 conventional schemas (11 wins, 0 losses each). Across Rounds 2–3 ($p = 0.21$) and CTU-14 (0.765 vs. 0.810), data-only RefSel and RefSel-base do not differ significantly, and with LLM evidence they converge (0.476 vs. 0.465, $p = 0.92$), showing that transfer is driven by Stage 1 candidates, Stage 3 CP-SAT decoding, and bounded LLM evidence. All-schema F1 in Round 3 is capped near 0.48 because 47 of 60 schemas are WikiDBs, where only 4% of label-mined keys target unique columns (capping WikiDBs F1 near 0.38).

Comparison with external baseline LLM-FK (P4): In Round 3, RefSel+LLM leads same-model LLM-FK by $+0.131$ on the 13 conventional schemas (0.819 vs. 0.688, Fig. 11(a)), $+0.092$ on unique-target keys ($p = 0.001$), and $+0.046$ across all 60 schemas (0.476 vs. 0.430; $+0.022$ on the 58 license-permitting schemas, 0.467 vs. 0.445, CI $[-0.012, 0.063]$, $p = 0.19$), with $7.5\times$ fewer calls ($1{,}769$ vs. $13{,}332$) and $< 1/20$ of the input tokens; without names, data-only RefSel and RefSel-base beat LLM-FK by $+0.092$ and $+0.082$ at zero calls (0.362 and 0.352 vs. 0.270, $p = 0.003, 0.013$). Round 2 shows the same advantage on all 41 schemas (0.493 vs. 0.411), the 11 conventional schemas (0.897 vs. 0.741), and the 39 license-permitting schemas (0.473 vs. 0.432), with $5.4\times$ fewer calls ($1{,}125$ vs. $6{,}026$) and $3.5\times$ higher precision (0.62 vs. 0.17).

6 Limitations

Data and ground truth: Stages 1–2 need populated tables: on sparse ERP demo installs, coincidental inclusions into tiny tables outnumber true keys and separating them needs application code or query logs. Declared keys are also incomplete: undeclared true references count as false positives for every method, as six such edges in the Round 1 audit show. Exact semi-joins limit Stage 1 on the widest schemas (full MusicBrainz and 912-table iDempiere hit the time/memory cap), where sketch-based filters could trade recall for speed.

Theory and risk control: The $1/\nu$ bound assumes exact relabeling symmetry (strictly indistinguishable targets), and Proposition 2 needs a frozen program and i.i.d. risk units. Our registered partitions do not establish independence, and Round 3 WikiDBs units share a Wikidata upstream across calibration and test. Under distribution shift (distractor tables, held-out named BIRD, or sparse ERP data), realized FDP exceeds α, calling for shift-aware calibration. Calibrating $\hat{\pi}_g$ does not calibrate c_e, which retains no more true edges than $\sigma(t_e)$ at fixed α.

Scope: Most Round 2–3 test schemas come from WikiDBs, whose label-mined keys rarely target unique columns. The seven external benchmarks were inspected during development, and LLM runs use two DeepSeek models, skip license-restricted schemas, and use (μ, B) tuned on an earlier scorer prototype.

7 Conclusion

Value inclusion is necessary for a foreign key, but it is weak evidence whose strength depends on the value domain and competing targets. A hypergeometric chance model makes this weakness measurable (Lemma 1), and an identifiability bound shows that, under relabeling symmetry, no method picks among ν indistinguishable targets above $1/\nu$ (Proposition 1); on BIRD and Spider, anonymous accuracy follows this $1/\nu$ limit.

RefSel turns a raw database into foreign keys with confidences through five components and a bounded LLM module (Remark 1). Across 355 internal databases, seven external benchmarks, and 14 CTU schemas, data-only RefSel beats published non-LLM baselines by $+4.2$ to $+18.9$ F1 points and anonymous LLM-FK by $+13.7$ points at zero calls; our Stage-2 ablation (RefSel-base) drives most of this lead, while s_e and stacking add $+1.2$ to $+4.7$ points on BIRD/Spider. On 110 unseen databases across three frozen rounds, bounded LLM evidence lifted both Stage 2 variants ($+18.9$ and $+23.1$ points on conventional schemas in Round 3), beating LLM-FK on conventional schemas (0.819

vs. 0.688), unique-target keys ($+9.2$ points), and anonymous schemas ($+9.2$ points) at $1/7.5$ of its calls and $< 1/20$ of its tokens while keeping realized FDP below $\alpha = 0.20$. Future work should calibrate retention under shift for sparse or distractor-heavy schemas, speed up Stage 1 on wide warehouses, and test recovered keys with query agents.

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